

Runtime User Messages



Message 1: My Tokyo departure is 9 a.m.

Message 2: The airline changed it to 11 a.m.

...

Trusted User-Level Event



target: tokyo_outbound_flight.departure_time
intent: replace_current_value
value: 11:00
scope: Tokyo trip
provenance: airline (confirmed)

Physical Memory (examples)

Tokyo departure time = 11:00 (active)

Tokyo departure time = 9:00 (superseded)

Seat preference = aisle

Dietary preference = vegetarian

Hotel preference = business hotel

...

PLACE

Factorized Governance Layer

Certifying Write-Side State (CSS)



User-Level Event
& Raw Utterance



Semantic State
Transition



Backend Update
& Verification



Maintain correct
persistent state

Authorizing Read-Side Influence (QASC)



Query Semantics
& Physical Readback



Deterministic
Authorization



Query Authorized
Support Construction



Expose only
query-eligible memory

Write
Governance
(CSS)

State
Realization

Read
Governance
(QASC)

Eligibility
Filtering

Update /
Readback

Authorized
Memories
(query-specific)

Persistent-Memory Backend
(Mem0)

Runtime User Query



Which is the current departure time,
and why?

Trusted Query Event



entity: tokyo_outbound_flight
predicate: departure_time
temporal view: current
request: provenance
scope: Tokyo trip

Query-Authorized Support (examples)

Tokyo departure time = 11:00 (authorized)

Flight change reason = airline (authorized)

Tokyo departure time = 9:00 (filtered)



Answer Model
(LLM)



Agent Response